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Inspiring Excellence

1. Project Title: Automated Water Management for Aquatic Systems Using Ultrasonic Sensors and Relay Control- "FishGuard"

Section: 5

Course: CSE360

Group: 5

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Abstract:

Our project, FishGuard, is designed to automate fish tanks, or more broadly, fish systems in aquatic environments, making it easier to monitor water levels, water quality, lighting, and feeding schedules. This system reduces the hassle of manual labor, ensuring efficient maintenance and a comfortable environment for the fish. Key features of this system include light control, automated feeding, real-time monitoring, and water level management. The lighting system adjusts the tank's lighting based on the time of day to simulate a natural day-night cycle. A servo motor is used as an automated feeder, ensuring fish are fed at regular intervals. An ultrasonic sensor monitors the water level, triggering the pump to drain and refill the tank up to the designated level. This pump is also connected to a turbidity sensor, which drains the water if it becomes too dirty. Communication between the Arduino microcontroller, sensors, and actuators is managed using I2C for connecting multiple sensors, SPI for high-speed communication with certain peripherals (such as a display or additional sensors), and UART for communication with external modules (e.g., Wi-Fi, Bluetooth) for remote monitoring and control. The goal of this project is to create a fully automated system that maintains the tank in optimal condition while minimizing the need for manual intervention. It also demonstrates how various components like sensors, motors, and microcontrollers can work together using advanced communication protocols to create a practical and effective solution.

2. Introduction

- **Problem Statement:** Maintaining the correct aquatic system in sources like fish tank or small fish farming or any aquatic creatures, is crucial for creatures' health, as well as feeding the fish on a regular interval and cleaning the tank are time-consuming and error-prone, creating a need for an automated system to adjust the water level without constant human intervention.

Objectives:

i) To develop a microcontroller-based system that automates the filling and draining of a fish tank, ensuring the water level remains within the optimal range.

ii) To interface an ultrasonic sensor with the microcontroller to measure the water level in real-time, allowing the system to respond based on the current water level.

iii) To control a DC motor through a relay module, managing water flow in and out of the tank, ensuring the water level is automatically maintained.

iv) To ensure reliable communication between the ultrasonic sensor, microcontroller, and motor relay using digital communication protocols such as I2C (for sensors requiring multiple connections), SPI (for high-speed peripherals), and UART (for communication with external modules such as Wi-Fi or Bluetooth).

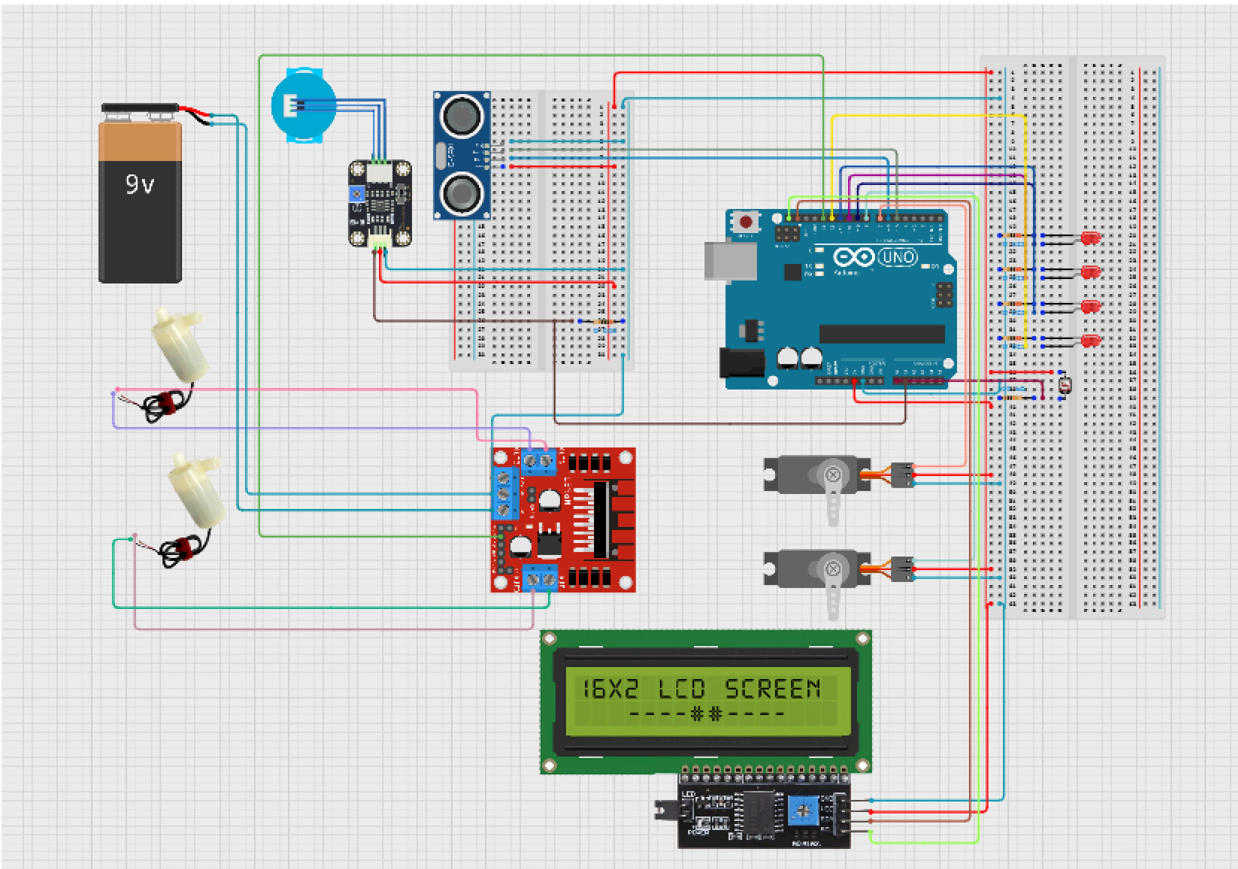
- **Significance**:
  In this technology-dependent world, our project, FishGuard, has various applications in In this technology-dependent world, our project, FishGuard, leverages interfacing technology to automate tasks such as water level monitoring, feeding schedules, and water refreshing, reducing human intervention and improving efficiency in both aquaculture and home fishkeeping. By integrating sensors, motors, and microcontrollers, it ensures real-time monitoring and enhances fish health, with potential applications in larger fish farms or smart home aquariums.

3. Interfacing Design

- **Interfacing Components**:
  - **Arduino Uno**:
    Acts as the central controller, processing data from the ultrasonic sensor and controlling the relay to activate the DC motor.
  - **Ultrasonic Sensor (HC-SR04)**:
    Measures the water level by emitting ultrasonic waves and calculating the time taken for the waves to return after hitting the water surface. This data is used to determine the distance and, subsequently, the water level in the tank.
  - **DC Motor (Water Pump)**:
    Used to pump water into the tank. The motor is controlled via a relay, which is activated or deactivated based on the water level.
  - **Relay (L298N, Driver motor)**:
    Acts as a switch to control the high-power DC motor using low-power control signals from the Arduino.
  - **Turbidity Sensor** (water quality monitoring)
  - **Light Dependent Resistor (LDR)** (ambient light measurement)
  - **Servo Motors** (automated feeder control)
  - **LEDs** (lighting control)

- ○ **Power Supply** (9V battery or 12V adapter)
- ○ **Breadboard and Wires** (for connections)
- ○ **LCD display**

System Diagram:



- **Interfacing Challenges**:
    - ○ **Signal Integrity**: We can cancel the noise from external sources that affect the readings of the ultrasonic sensor. Inorder to control this, the sensor wiring should be properly protected.
    - ○ **Compatibility Issues**: We have to check that the relay module works smoothly with the Arduino and motor requires checking voltage and current ratings. We can use simple transistors like 3 terminal ones such as : mosfets,bjt may to drive the relay.

    - **Waterproofing**: Components like the ultrasonic sensor and relay module should be guarded from water exposure, we were not able to work extensively in this field , we just positioned the components carefully , in order to prevent them from water exposure.

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4. Communication Protocols

- **Protocol Selection**:
  The project uses **digital I/O communication** between the Arduino and the components.
    - **Ultrasonic Sensor**: Communicates with Arduino through **digital pins** using the **pulse-width modulation (PWM)** technique to measure distance.
    - **Relay Module**: Uses simple **digital signals** from the Arduino to control the motor (HIGH to activate, LOW to deactivate).
- **Protocol Justification**:
    - **Ultrasonic Sensor (HC-SR04)** uses digital communication, which is fast and ideal for the short-range measurement of water levels. Since the data rate and latency requirements are low, this protocol is sufficient.
    - The **relay** uses a **digital signal** to control the motor. Since the relay switches a simple high/low signal, digital communication is optimal for turning the pump on/off based on sensor input.

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5. Implementation Plan

- **Methodology**:
  The system will first read the water level via the ultrasonic sensor. Based on the measured distance, it will:
    - Start filling the tank by activating the relay and the motor if the water level is below 50%.

- Stop the pump when the water level reaches 90%. To measure the water level distance we will use an ultrasonic sensor combined with the motors as the first motor would turn on and off based on the readings of the ultrasonic sensor. The Arduino code will process sensor data and use the l298n motor driver relay to control the motor.

- Here we have used the l298n motor driver module to power our DC motors which would be used to pump out and pump in water from the Container. Here the motor driver is connected to the power supply , a 9V battery as power supply is used. Here the ground of the battery and the ground of the Motor driver circuit should be connected to a common point . The motor driver circuit has input pins which would be connected to the Digital pins of the arduino . Here we will be using 2 motors for pumping in and pumping out the water from the container thus 2 input pins of the motor driver circuit would be used . Here two 9V batteries are also used as power supply for driving two DC motors.

- The second motor would pump out the water as the first motor pumps the water in. The second motor would pump out the water based on the readings on the turbidity sensor. The turbidity sensor indicates the haziness and the gloominess of the water. If the water is too hazy then the turbidity sensor gives higher readings or values which would indicate that it is time to change the water and thus our motor would be activated to pump out the water.

- After the motors there is an automatic feeder system controlled by servo motors here we will be using 2 servo motors. The first servo motor is connected to the lid of the container , the container carries the food for the fishes , under the lid of the container there is a tray (which is also controlled by the servo motors). When the lid opens meaning our 1st servo motor goes from 90 degrees to 0 degrees then immediately the second servo motor which controls the tray also moves from 90 degrees to 135 degrees allowing the food to fall into the tray. After when the food falls into the tray the lid immediately closes and then the second servo motor moves from 135 degrees to 180 degrees allowing the food to fall into the container from the tray thus making the cycle repeat again after certain intervals.

- There are leds which would control the intensity of light around the container . The led is controlled by using a light dependent resistor. Usually in average lighting conditions the only 2 out of the 4 LEDs would light up. If the intensity of light in the surrounding is low then all 4 of the LED would light up and if the intensity of light in the surrounding is high then all of the LED would be turned off. It is basically implemented by using a voltage divider circuit connecting 4 LEDs with a resistor and a Light dependent resistor along with it , which would measure the intensity of light.

- There is an LCD display as well which would give warning signals like , when the light is low and when the light is high , when the water is normal and when the water is hazy, when to feed the fish and when to not , this kind of texts would be displayed.

Arduino Code:

The Arduino code for controlling the pump based on the water level is as follows:

```
Initialize the following pins:
  turbiditySensorPin = A0 (Analog Pin for Turbidity Sensor)
  motorPin = 9 (Digital Pin for Motor Control)
  ldrPin = A0 (Analog Pin for LDR Sensor)
  ledPins[4] = {9, 10, 11, 12} (Digital Pins for LEDs)
  lidServoPin = 6 (Pin for Servo controlling lid)
  trayServoPin = 5 (Pin for Servo controlling tray)

Set initial servo positions:
  lidServo.write(90) // Lid closed
trayServo.write(90) // Tray in initial position

Define thresholds:
  HIGH_LIGHT_THRESHOLD = 800
  LOW_LIGHT_THRESHOLD = 300
  turbidityThreshold = 90

Start Serial Communication for debugging
```

```
Initialize the feeder state machine:
  FeederState = INITIAL_WAIT

Forever loop:

  Read turbidity sensor value:
    turbidityValue = analogRead(turbiditySensorPin)

  Check turbidity value against threshold:
    if turbidityValue > turbidityThreshold:
      Turn motor ON
else:
      Turn off Switch motor

  Read the value of LDR:
    ldrValue = analogRead(ldrPin)

  Check if LDR value has changed:
    if ldrValue < LOW_LIGHT_THRESHOLD:
      Turn on LEDs
    else if ldrValue > HIGH_LIGHT_THRESHOLD:
      Turn off LEDs
    else:
Control LEDs by range

  Update servo state for feeder:
    currentTime = millis()
    switch FeederState:
      case INITIAL_WAIT:
        if currentTime - lastStateChange >= 10000:
          Open lid
Tilt tray
          FeederState = OPEN_LID_AND_TILT
          lastStateChange = currentTime
      case OPEN_LID_AND_TILT:
        if currentTime - lastStateChange > 5000:
          Close lid
Set FeederState = CLOSE_LID
```

```
          Set lastStateChange = currentTime
      case CLOSE_LID:
        Set FeederState = TRAY_WAIT
        Set lastStateChange = currentTime
      case TRAY_WAIT:
if currentTime - lastStateChange >= 2000:
          Tilt tray to drop food
          Set FeederState to DROP_FOOD
          Set lastStateChange = currentTime
      case DROP_FOOD:
if currentTime - lastStateChange >= 12000:
          Set tray to Reset state
          Set FeederState to RESET_POSITION
          Set lastStateChange = currentTime
      case RESET_POSITION:
        Set FeederState to INITIAL_WAIT
Set lastStateChange = currentTime

 Wait for some time before the loop is repeated
```

- **Expected Outcomes**:
    - The system will automatically fill the tank when the water level drops below 50% and stop when it reaches 90%.
    - The pump will be controlled based on real-time measurements from the ultrasonic sensor.

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6. Future Work and Potential Applications

- **Future Improvements**:
    - Add more sensors for monitoring water quality, like pH sensor and temperature sensors like DS18B20 or TMP102 with an I2C interface to understand the weather and condition of the water.
    - Integrate the system with a mobile app to monitor and control the fish tank remotely.

    - Use of powerful and large-capacity components for architectural artificial water ponds in commercial buildings or parks, and even for fish farming.

- **Applications**:

    - This project can be applied in home aquariums to automate water management, reducing the need for manual intervention. In scenarios where users are absent or far from home, the system ensures that their fish tanks continue to operate efficiently without the need for direct supervision.
    - Aquaculture operations can benefit from a scalable version of this system to manage water levels, ensuring optimal conditions for fish health and improving overall efficiency in fish farming.
    - The project demonstrates practical applications of interfacing components like sensors, actuators, and microcontrollers, while also utilizing communication protocols for seamless operation and remote monitoring, making it ideal for real-world scenarios in both small-scale and large-scale fishkeeping.

7. References

- Melvin, G. D., Dadswell, M. J., & Williams, P. J. (1983). Effects of turbidity and the temporal and spatial utilization of the inner Bay of Fundy by American shad (Alosa sapidissima). Canadian Journal of Fisheries and Aquatic Sciences, 40(3), 382–390. https://doi.org/10.1139/f83-056

- Syukriyadin, S., Syahrizal, S., Mansur, G., & Ramadhan, H. P. (2018). Permanent magnet DC motor control by using Arduino and motor drive module BTS7960. IOP Conference Series: Materials Science and Engineering, 352(1), 012023. https://doi.org/10.1088/1757-899X/352/1/012023

Conclusion

The automated aquarium setup successfully utilizes an ultrasonic sensor, a relay module, and a DC motor to control water automatically such that the tank is kept at the ideal level of water all the time. This setup reduces manual interference and ensures a stable and healthy condition for aquatic life, highlighting real-world applications of microcontroller interfacing and sensor communication.